

MSE 312 — Combined Board Soldering Instructions

Solder from scratch on a solderable breadboard: SN7400 3-gate motor logic + TB6568KQ driver + 4N26/MOSFET solenoid driver (12–24 V), with all MCU-protection add-ons. Read fully before starting.

Safety & prep

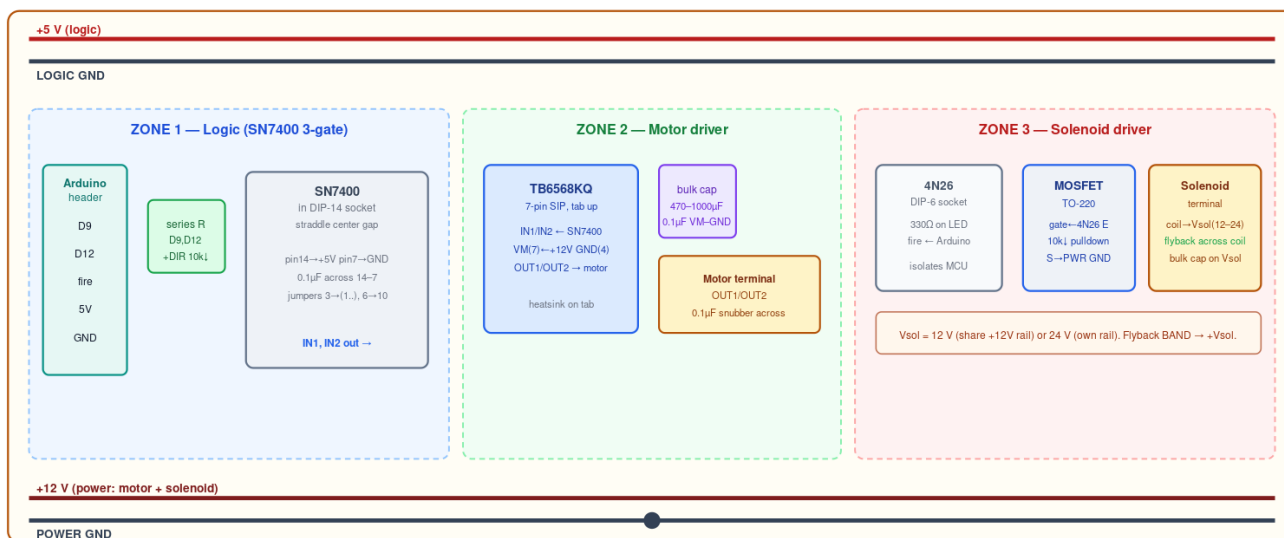
- Work with power OFF and everything unplugged. ESD: touch ground before handling the SN7400 / 4N26.
- Temperature-controlled iron ~330–350°C, fine tip, thin (0.6–0.8 mm) rosin-core solder, flux, flush cutters, magnifier, multimeter.
- Do a DRY placement first — lay every part on the board and confirm orientation before any solder flows.
- Use IC SOCKETS for the SN7400 and 4N26 — never solder those chips directly (heat + lets you swap after the last board died).

Bill of materials

Qty	Part	Notes
1	Solderable breadboard	bus rails top & bottom, center-gap rows
1	SN7400 + DIP-14 socket	quad NAND, 3 gates used
1	TB6568KQ	7-pin SIP H-bridge; small heatsink on tab
1	4N26 + DIP-6 socket	optocoupler (solenoid isolation)
1	N-MOSFET (TO-220)	logic-level pref.; $V_{ds} \geq 30\text{ V}$, $\geq$ coil current
1	Flyback diode	1N4007 or Schottky, $\geq$ coil current
1	Electrolytic 470–1000 μF , $\geq 25\text{ V}$	motor VM bulk; + a solenoid-rail bulk cap
4–5	Ceramic 0.1 μF	IC decoupling, motor snubber, cap-pairing
~4	Resistors	2× series 220 Ω –1k (D9,D12); 10k DIR pulldown; 330 Ω LED; 10k gate pulldown
3	Screw terminal blocks	12 V in, motor out, solenoid out
1	Male header strip	Arduino jumper connections
-	Hookup wire	solid 22–24 AWG for jumpers; heavier for power

Board layout

Solderable-breadboard layout (placement map, not to scale)



STAR GROUND: join LOGIC GND + POWER GND + supply – at ONE point here
Signals flow left→right. Power rails run along top (5V) and bottom (12V). Opto keeps the fire signal isolated from the solenoid switching node.

Signals flow left→right; +5 V logic on the top rail, +12 V power on the bottom rail; star ground at center.

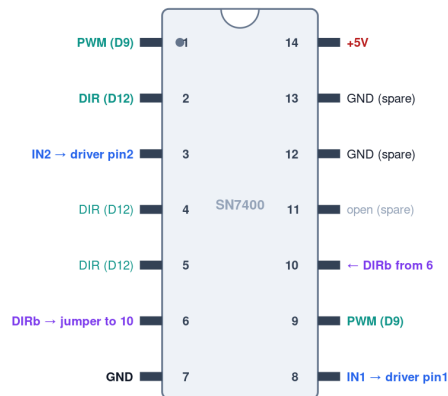
Solder order (low & heat-sensitive first)

1. Solder the two IC SOCKETS (SN7400 DIP-14, 4N26 DIP-6) straddling the center gap. Leave ICs OUT for now.
2. Solder all RESISTORS (series R on D9 & D12, 10k DIR pulldown, 330Ω LED, 10k gate pulldown).
3. Solder all 0.1 μF CERAMICS (one across each IC's Vcc–GND, one motor snubber, one paired with each bulk cap). Non-polar — any way round.
4. Solder the FLYBACK DIODE — band (cathode) toward +Vsol. Polarity critical.
5. Solder the ELECTROLYTIC bulk caps — LONG leg = +, striped short leg = -. Polarity critical.
6. Solder the TB6568KQ (tab up, heatsink) and the MOSFET (TO-220).
7. Solder the TERMINAL BLOCKS and Arduino HEADER.
8. Run the POWER RAILS, then the signal JUMPERS per the tables. Do grounds / star point LAST.
9. Inspect under magnifier for bridges; buzz out continuity; THEN insert the ICs (mind the notch).

Zone 1 wiring — SN7400 (3-gate logic)

SN7400 pin map — 3-gate config (D9 PWM · D12 DIR)

Top view, notch up. IN2 = NAND(PWM,DIR), DIRb = NOT DIR, IN1 = NAND(PWM,DIRb)



PWM (D9) → pin1 & pin9 · DIR (D12) → pin2, 4 & 5 · spare gate 4 inputs (12,13) tied to GND

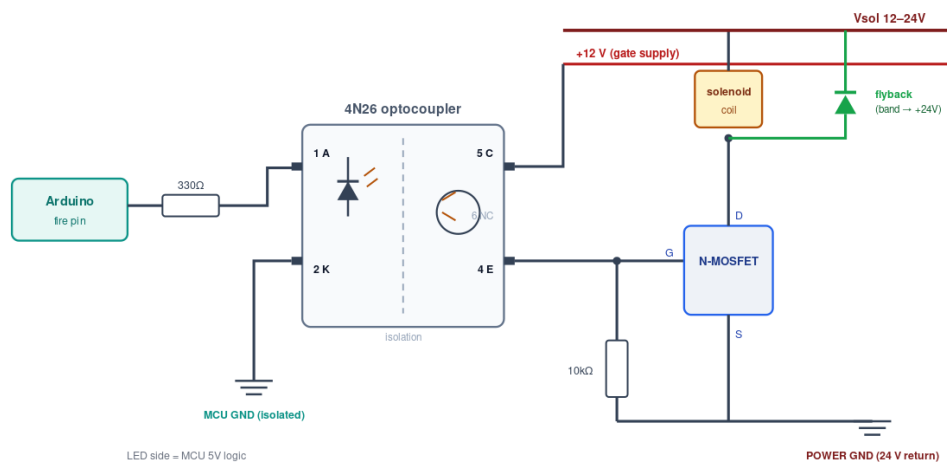
From	To	Purpose
Arduino D9 (PWM)	series R → SN7400 pin 1 AND pin 9	PWM to both NANDs
Arduino D12 (DIR)	series R → SN7400 pins 2, 4 & 5	direction + inverter input
D12 line	10 kΩ → GND	pull-down (defaults CW if wire lost)
SN7400 pin 6 (DIRb)	SN7400 pin 10	inverted DIR into gate 3
SN7400 pin 3	TB6568KQ pin 2 (IN2)	IN2
SN7400 pin 8	TB6568KQ pin 1 (IN1)	IN1
SN7400 pins 12 & 13	GND	tie off spare gate 4
SN7400 pin 14 / pin 7	+5V / GND (0.1μF across)	logic power + decoupling

Zone 2 wiring — TB6568KQ motor driver

From	To	Purpose
TB6568KQ pin 7 (VM)	+12 V rail	motor supply
TB6568KQ pin 4 (GND)	power GND (star)	ground
TB6568KQ pins 3,5 (OUT1/2)	motor terminal block	to motor
Bulk cap 470–1000µF ■ 0.1µF	across VM ↔ GND at the chip	absorb inrush/reversal spikes
0.1 µF snubber	across motor terminals (OUT1↔OUT2)	kill brush EMI
pin 6	(leave open)	N.C.

Zone 3 wiring — 4N26 / MOSFET solenoid driver (12–24 V)

Solenoid driver: Arduino → 4N26 opto → N-MOSFET → 24 V coil (flyback)



From	To	Purpose
Arduino fire pin	330Ω → 4N26 pin 1 (A)	LED drive
4N26 pin 2 (K)	logic GND	LED return
4N26 pin 5 (C)	+Vsol (gate supply)	phototransistor collector
4N26 pin 4 (E)	MOSFET gate	gate drive
MOSFET gate	10 kΩ → power GND	pull-down (default OFF)
MOSFET source	power GND	low-side switch
MOSFET drain	solenoid coil (–)	switched side
Solenoid coil (+)	+Vsol (12 V share rail, or 24 V own rail)	coil supply
Flyback diode	across coil, BAND → +Vsol	clamp turn-off spike (mandatory)
Bulk cap	across +Vsol ↔ power GND	absorb firing pulse

Grounding — do this right or it bites you

Bring **Arduino GND**, **SN7400 pin 7**, **TB6568KQ pin 4**, the **12 V supply** –, and **both bulk-cap** – to ONE star point (near the 12 V terminal). Don't daisy-chain the motor return through the logic ground. The 4N26 keeps the Arduino's fire pin off the solenoid switching node — but the solenoid coil still returns to power GND, so the star point matters. This is the exact fault (ground bounce + inductive spike) that killed the previous board.

Bring-up test — staged, power added one stage at a time

Stage	Do	Pass criteria
A. Visual	magnifier sweep	no solder bridges; elec-cap + and flyback band correct; IC notches match
B. Continuity (ICs OUT, power off)	buzz +5V–GND, +12V–GND, +Vsol–GND; verify each net vs tables	no rail-to-GND short; nets match
C. Logic only (5 V / USB)	insert SN7400; toggle D9/D12; scope IN1/IN2	IN1/IN2 follow the truth table; no heat on SN7400
D. + Motor (12 V)	low duty; sweep direction	motor turns both ways; VM stays up; no Arduino reset
E. + Solenoid (Vsol)	insert 4N26; pulse fire pin briefly	solenoid clicks; no MCU glitch/reset; MOSFET stays cool

If something's wrong

- SN7400 or 4N26 gets HOT → power off; check orientation/Vcc–GND not reversed and no output short.
- Motor won't move / wrong way → recheck IN1/IN2 wiring; swap motor leads for direction (don't rewire logic).
- Arduino resets when solenoid or motor fires → grounding/decoupling; verify star ground, bulk caps, flyback.
- Any pin reads ~0.3 V on HIGH → transient got through; stop and recheck series resistors + grounds before continuing.